Will Walters

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Education

Purdue University College of Engineering

- Bachelor of Science in Aeronautical and Astronautical Engineering (AAE), GPA: 3.90/4.00 — August 2022 - May 2026
- Master of Science with Thesis in AAE — August 2026 - May 2028

Skills

- **Programming:** MATLAB/Simulink, Python, C/C++, Java
- **Engineering Software:** GMAT, FreeFlyer, AutoCAD, XFLR5, OpenRocket
- **Technical Skills:** Orbital Mechanics, Spacecraft Dynamics, Optimal Control, State Estimation

Professional Experience

Manufacturing Engineering Intern | Subaru of Indiana Automotive Inc. — June - August 2025

- Conducted feasibility studies for plant energy reduction by analyzing facility layouts for energy efficiency and leveraging energy capture for existing systems. Used plant data for energy usage and system performance to inform results.
- Led an inverter installation and network connection sub-project for new solar panels by writing and sending RFQs, performing walkthroughs with vendors and other teams at the plant, and submitting a Purchase Order.

Research Assistant | Purdue Manufacturing and Materials Research Labs — June - August 2026

- Evaluated flexible and rigid Printed Circuit Board (PCB) technology and tested thick-flim printing and PCB assembly with a Direct Ink Writing system.
- Used the principles of expeditionary manufacturing to lead the initial stages of a mobile workshop project for PCB fabrication, serving as the point of contact with experts on Conex containers and isolation vibration components.

Graduate Teaching Assistant | Purdue Mathematics Department — August 2026 - Current

- Leading two Calculus 1 recitation sections twice a week, assisting students with homework and lecture questions, and administering and grading quizzes.

Project Experience

Space Tether Rendezvous in Cislunar Space — January 2026 - Current

- Worked with a partner to develop a problem framework and implement trajectory optimization algorithms for rendezvous with a rotating space tether in a Lyapunov orbit about the Earth-Moon L_1 point
- Own the research and implementation of integrating a free final time and adaptive mesh solution with the existing direct parameterization and Sequential Convex Programming algorithms developed.

Orbital Debris Removal System | GNC Lead | Ground Systems Deputy — January - May 2026

- Led the development of a Guidance, Navigation, and Control (GNC) sub-system in an eleven-person multidisciplinary team formulating a mission to remove large defunct satellites in Sun-synchronous orbits.
- Used orbit simulation results and sensor performance to inform critical decisions about mission design and architecture, improve mission operational robustness, and ensure seamless integration between the GNC sub-system and others.

Java Social Media Application | User Profile and Client Development Lead — March - April 2024

- Collaborated in a four-person team to design a multi-thread social media application with a graphical interface.
- Responsible for the development of the user profile system, ensuring coherent communication between the client and server, and contributed to the Graphical User Interface (GUI) implementation.

Perseverance Rover Sample Collection Mission — August - November 2023

- Designed a future dual-vehicle mission concept for retrieving samples collected by NASA's Perseverance rover.
- Performed analysis on stakeholder needs, mission requirements, risks, and mission parameters such as timeline, mission architecture, and payload sizing.

Extracurricular Activities

Purdue SLING Research Club | Controls Team — February 2025 - Current

- Work in a small group to model and analyze control and estimation of a spacecraft and tether in cislunar space.
- Apply dynamics and control theory to critically analyze mission scenarios while leveraging existing research.

Purdue Club Lacrosse — September 2023 - Current

- Compete with a large team in regional and national competitions in the Men's Collegiate Lacrosse Association (MCLA).